João Pedro Sousa Gaspar



Aerospace Engineering graduate from Instituto Superior Técnico, with a dual background in Electronics Engineering and a specialization in Aerodynamics and Propulsion. Passionate about computational fluid dynamics, reusable technology, and the future of sustainable world. Hardworking and dedicated, I bring hands-on experience combined with a strong motivation to keep learning and contribute meaningfully to cutting-edge engineering challenges.

Education

Master's degree in Aerospace Engineering
Sep 2022 – Jun 2025

Instituto Superior Técnico

- Aerodynamics and Propulsion specialization
- **Key subjects:** Incompressible and Compressible Aerodynamics, Helicopters, Computational Fluid Dynamics, Aircraft Optimization, Electric Propulsion for Aircraft, Project Management
- **Thesis Subject:** *Computational Investigation of Spacecraft Recovery via Rotary Wings: A Study on Autorotation*

Bachelor's degree in Electronics Engineering
Sep 2019 – Jun 2022

Instituto Superior Técnico

- **Key subjects:** Digital Systems, Communications Systems, Embedded Systems, Instrumentation and Signal Acquisition, Micro and Nanoelectronics, Computer Networks, Fundamentals of Automatic Control, Electronic Devices, Management

Experience

Self-Defense Instructor
Sep 2022 – Present

Krav Maga Global

- Planned and led self-defense classes for individuals and groups, ensuring safety and engagement.
- Provided personalized guidance to students, adapting techniques to different skill levels.
- **Developed Skills:** leadership, communication, teaching, adaptability, attention to safety

Scholarship
Nov 2021 – Jan 2025

ROB9-16 & Project "Oeiras Educa"

- Led STEM workshops and activities for thousands of children under the ROB9-16 project at Instituto Superior Técnico.
- Collaborated with Câmara Municipal de Oeiras to develop educational programmes, lectures, and student projects.
- **Developed Skills:** work with children, public speaking, teaching, communication

Técnico Fuel Cell
Jul 2022 – Apr 2024

Electrical Engineer

- Development of a telemetry system for a hydrogen powered car for Shell Eco-Marathon competition
- Responsible for the electronics and car assembly workshops, for car development, service and maintenance
- **Developed Skills:** electronics, PCB design, circuits and systems design, laboratory work, management

Volunteering

Mentor
Sep 2020 – Jul 2022

NAPE - IST

- Welcome and integrate new students to the school

Project "Salas de Estudo Online"
Sep 2021 – Jan 2022

Núcleo de Desenvolvimento Académico - IST

- Providing personalized classes for the first years students helping with projects and examinations

Certifications

KMG Civil Instructor Course (190h) - 2023

Advanced Certificate in English - 2019

First Certificate in English - 2018

Hard Skills

Programming Languages: Matlab, Python, C

Software: Microsoft Office, Simulink, LabView, Solidworks, STARCCM+

Driving license for category B

E-mail | LinkedIn | Webpage